

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Semestertoets 2

VRAESTEL

Kopiereg voorbehou

Module EBN111

Elektrisiteit en Elektronika

Mei 2017

Department of Electrical, Electronic and
Computer Engineering

Semester Test 2

QUESTION PAPER

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Module EBN111

Electricity and Electronics

May 2017



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

MEMO

Maksimum punte: <i>Maximum marks:</i>	46	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 min + 10 min to complete OCR form <i>90 min + 10 min om OKH vorm te voltooi</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		17 + OCR	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word.
All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
Take careful consideration of the instructions on the OCR form.
4. Maak gebruik van 5 beduidende syfers in antwoorde bestaande uit irrasionele getalle.
Make use of 5 significant digits for all answers involving irrational numbers.

Dosent(e): <i>Lecturer(s):</i>	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 before the start of the test.

Do step 1 in the first 90 minutes of the test

Do step 2 within the following 10 minutes.

STEP 0 (– 5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2017EBN111S2**
- The correct units for each question are shown on the PRACTICE ANSWER SHEET.
- Use a **mechanical pencil** to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	–	1	.	4	5					m	A								
02	3	4	0							V									
03	1	5	.	2															
04	-	1								Ω									

INSTRUKSIES

Doen stap 0 voor die begin van die toets.

Doen stap 1 in die eerste 90 minute van die toets.

Doen stap 2 binne die daaropvolgende 10 minute.

STAP 0 (– 5 min to 0 min): Voltooi die voorblad van die OKH vorm.

STAP 1 (0 min to 90 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (90 min to 100 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.

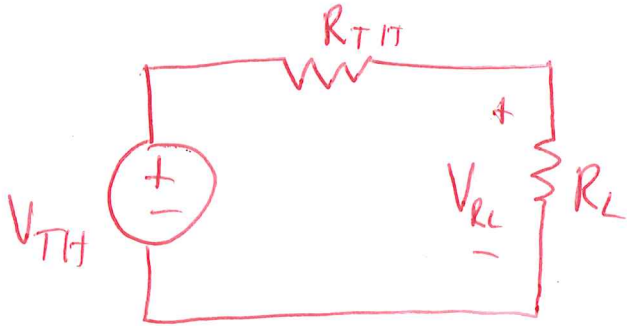
- Die “Assessment ID” is **2017EBN111S2**
- Die korrekte eenhede vir al die vrae in die toets word reeds op die OEFEN ANTWOORDBLAD aangedui.
- Gebruik ‘n meganiese potlood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak eerde gebruik van punte.

Voorbeelde

01	–	1	.	4	5					m	A								
02	3	4	0							V									
03	1	5	.	2															
04	-	1								Ω									

SECTION / AFDELING 1 [10]

Question 1 & 2 [2]		Vraag 1 & 2 [2]	
Assume that a Thevenin equivalent circuit is connected to a load resistor R_L and that maximum power p_{max} is delivered to the load. If the voltage across the R_L is 20 V and $p_{max} = 800$ mW, find the Thevenin equivalent voltage V_{TH} and resistance R_{TH}		'n Thevenin ekwivalente kringbaan word gekoppel aan 'n lasweerstand R_L , en lewer maksimum drywing p_{max} aan die las. Indien daar 'n spanning van 20 V oor R_L is en $p_{max} = 800$ mW, bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} .	
01	V_{TH}	01	V_{TH}
02	R_{TH}	02	R_{TH}



$$V_{RL} = \frac{R_L}{R_L + R_{TH}} V_{TH}$$

$$\therefore V_{TH} = \frac{R_L + R_{TH}}{R_L} V_L$$

$$\text{but } R_L = R_{TH}$$

$$\therefore V_{TH} = \frac{2R_L}{R_L} V_{RL}$$

$$= 2V_{RL}$$

$$= 2(20)$$

$$V_{TH} = 40V$$

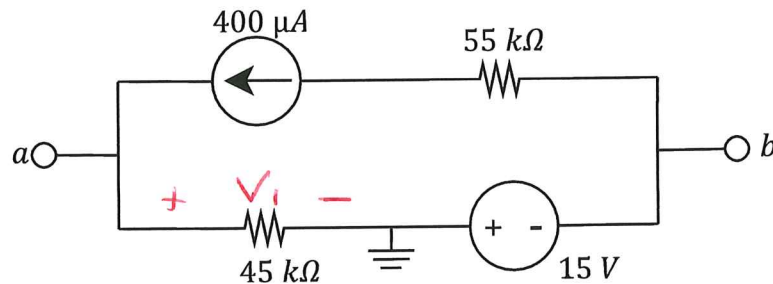
$$P_{max} = \frac{V_{th}^2}{4R_{TH}}$$

$$\therefore R_{TH} = \frac{V_{th}^2}{4P_{max}}$$

$$= \frac{40^2}{4(800 \times 10^{-3})}$$

$$R_{TH} = 500 \Omega$$

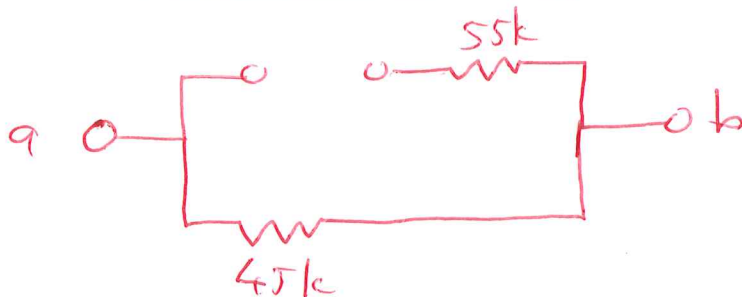
Question 3 & 4 [4]		Vraag 3 & 4 [4]	
Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} w.r.t. terminals a - b .		Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} met verwysing tot terminale a en b .	
03	V_{TH}	03	V_{TH}
04	R_{TH}	04	R_{TH}



$$V_1 = 45k(400\mu) = 18V$$

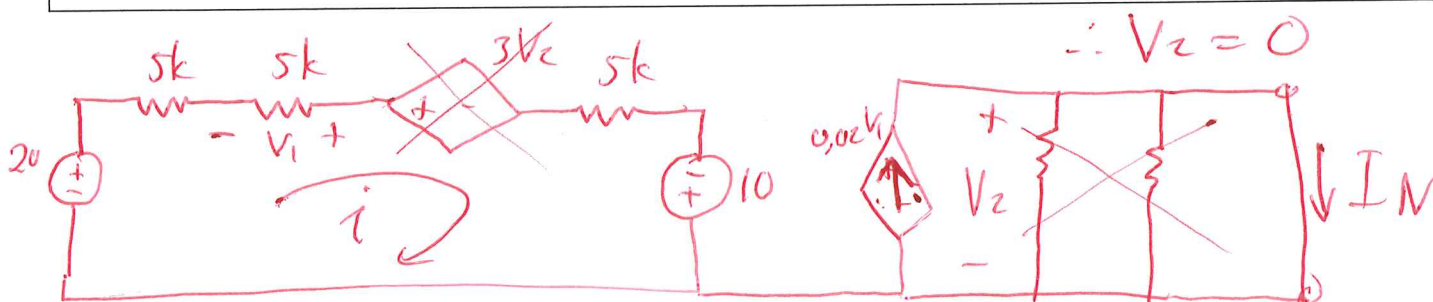
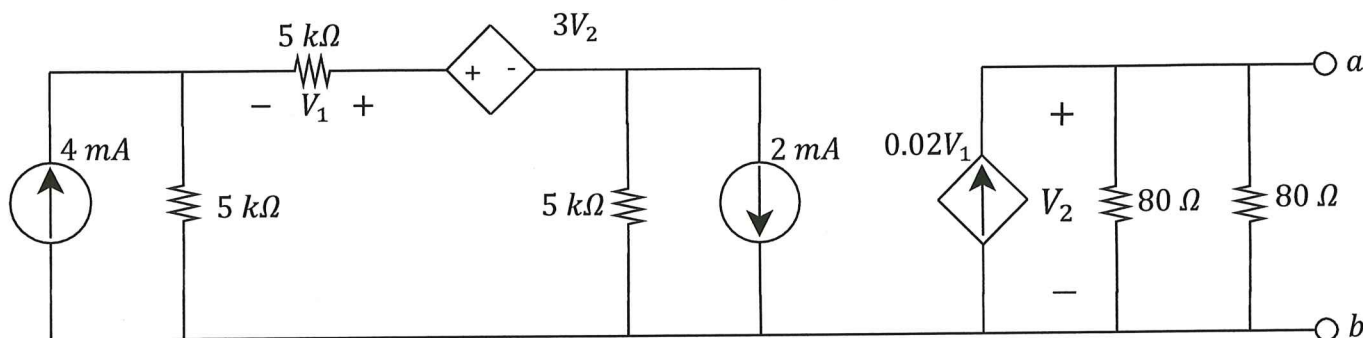
$$\therefore V_{TH} = V_1 + 15 = 18 + 15$$

$$V_{TH} = 33V$$



$$\therefore R_{TH} = 45k\Omega$$

Question 5 & 6 [4]		Vraag 5 & 6 [6]	
Determine the Norton equivalent current I_N and resistance R_N w.r.t. terminals a - b .		Bepaal die Norton ekwivalente stroom I_N en weerstand R_N met verwysing tot terminale a en b .	
05	I_N	05	I_N
06	R_N	06	R_N

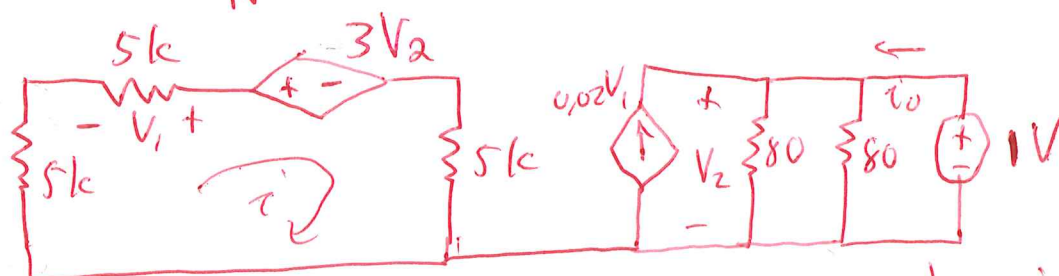


$$-20 + 15k i - 10 = 0$$

$$\therefore i = 2 \text{ mA}$$

$$\therefore V_1 = -(2 \text{ m})(5k) = -10 \text{ V}$$

$$\therefore I_N = 0,02(-10) = -200 \text{ mA}$$



$$15k i + 3(1) = 0$$

$$\therefore i = -0,0002 \text{ A}$$

$$\therefore V_1 = -(-0,0002)(5k) = 1 \text{ V}$$

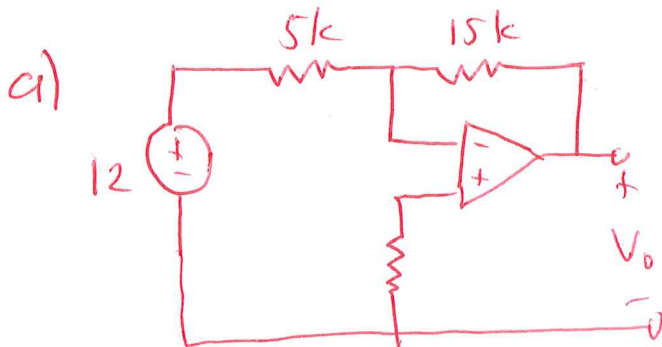
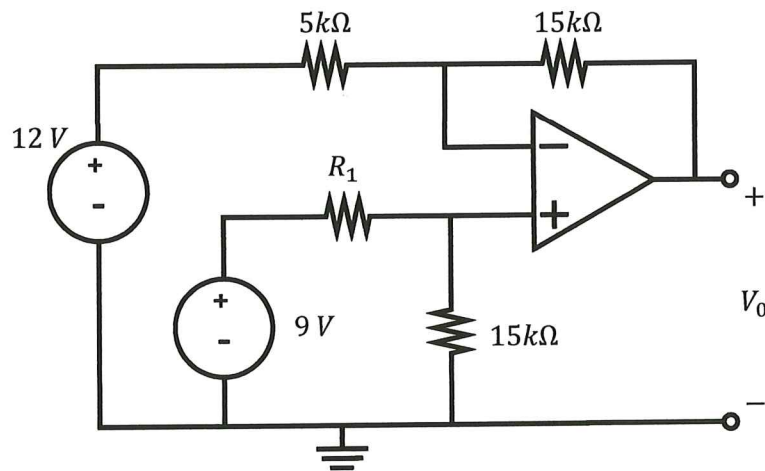
$$V_2 = 1 \text{ V}$$

$$i_0 = \frac{2}{80} - 0,02 = \frac{1}{200}$$

$$\therefore R_N = \frac{1}{i_0} = 200 \Omega$$

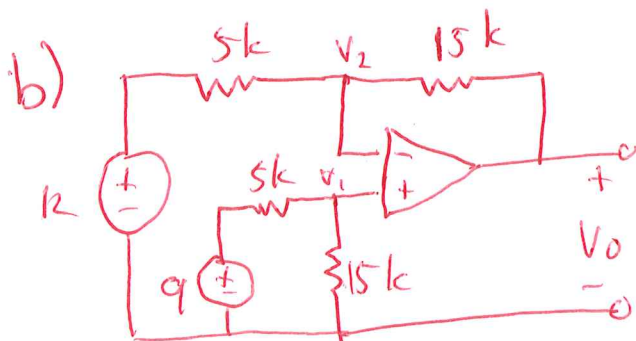
SECTION / AFDELING 2 [16]

Question 7 & 8 [4]		Vraag 7 & 8 [4]	
07	If $R_1 \rightarrow \infty \Omega$, calculate V_0 .	07	Indien $R_1 \rightarrow \infty \Omega$, bereken V_0 .
08	If $R_1 = 5 k\Omega$, calculate V_0 .	08	Indien $R_1 = 5 k\Omega$, bereken V_0 .



$$V_0 = -\frac{15}{5} \times 12$$

$$V_0 = -36 V$$



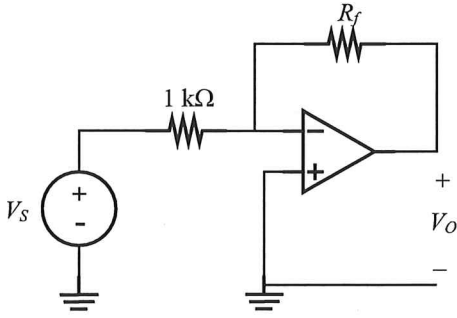
$$V_1 = V_2 = \frac{15k}{5k + 15k} (9)$$

$$= 6,75 V$$

$$\frac{V_2 - 12}{5k} + \frac{V_2 - V_0}{15k} = 0$$

$$\times 15k: (6,75 - 12)(3) + 6,75 - V_0 = 0$$

$$\therefore V_0 = -9 V$$

Question 9 & 10 [4]				Vraag 9 & 10 [4]			
The range of the output voltage V_O is: $-10 \leq V_O \leq 1$ [V]				Die bereik van die uitset spanning V_O is: $-10 \leq V_O \leq 1$ [V]			
The range of the input voltage V_S is: $-1 \leq V_S \leq 5$ [V]				Die bereik van die inset spanning V_S is: $-1 \leq V_S \leq 5$ [V]			
Determine the minimum and maximum value of R_f which will satisfy above constraints, i.e. calculate a and b for the equation below: $a \leq R_f \leq b$				Bepaal die minimum en maksimum waarde van R_f wat bostaande kriteria sal bevredig, m.a.w. bereken a en b vir onderstaande vergelyking: $a \leq R_f \leq b$			
Remember, R_f may not be negative.				Onthou, R_f mag nie negatief wees nie.			
09	a	10	b	09	a	10	b
 inverter							

$$V_O = -\frac{R_f}{1k} (V_S)$$

$$-10 \leq V_O \leq 1$$

$$-10 \leq -\frac{R_f}{1k} (V_S) \leq 1$$

$$-10k \leq -R_f V_S \leq 1k$$

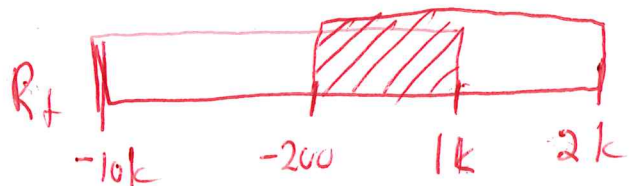
$$\text{At } V_S = -1:$$

$$-10k \leq R_f \leq 1k$$

$$\text{At } V_S = 5:$$

$$-10k \leq -5R_f \leq 1k$$

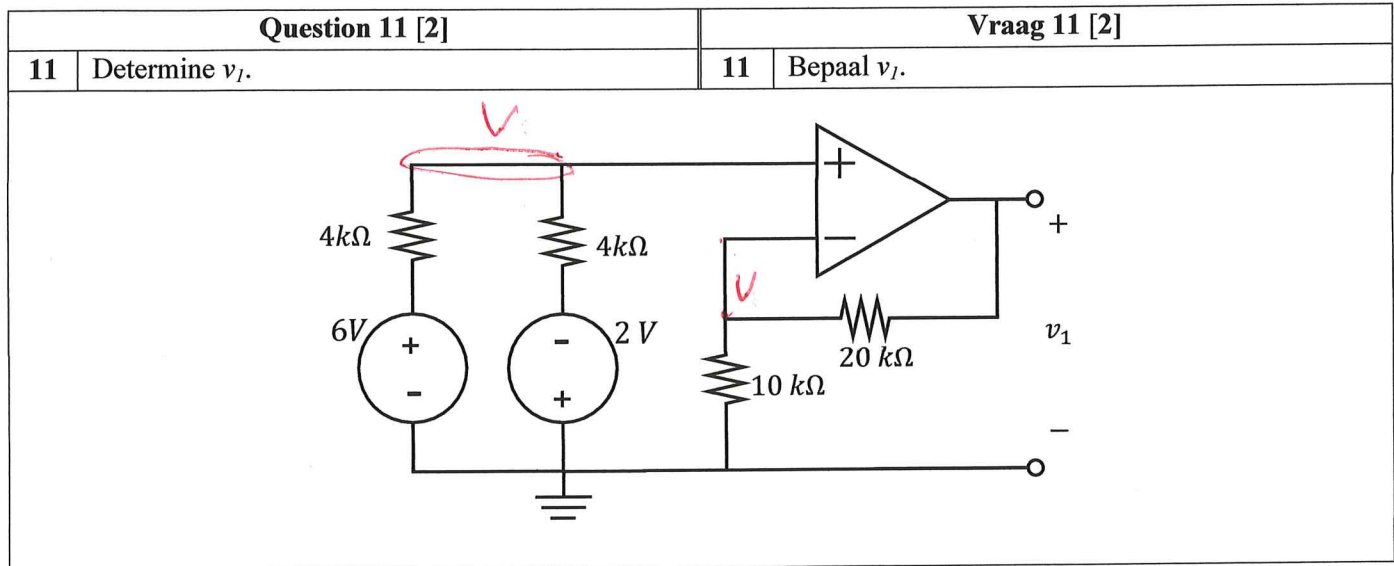
$$-200 \leq R_f \leq 2k$$



$$\therefore -200 \leq R_f \leq 1k$$

$$\text{but } R_f \geq 0$$

$$\boxed{0 \leq R_f \leq 1k}$$



$$\frac{V - 6}{4k} + \frac{V + 2}{4k} = 0$$

$$\times 4k: 2V - 6 + 2 = 0$$

$$\therefore V = 2V$$

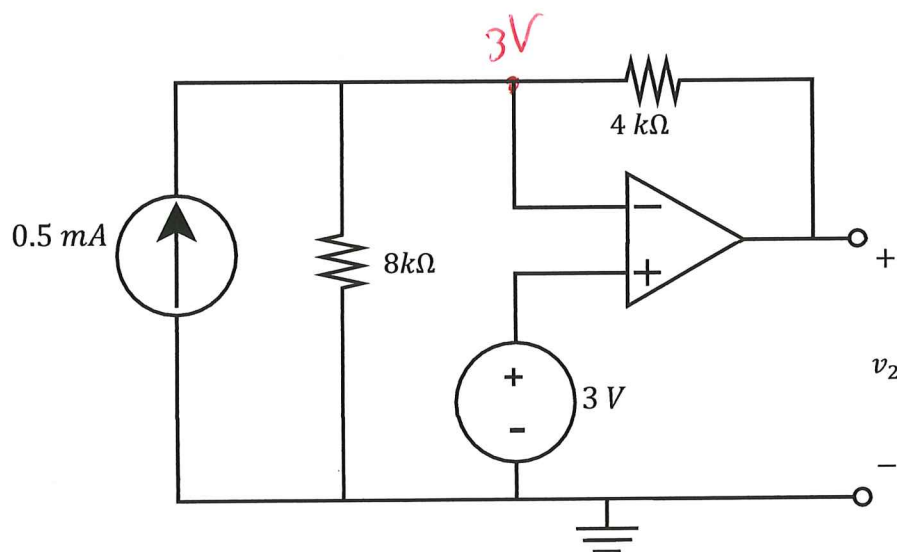
$$\frac{V}{10k} + \frac{V - V_1}{20k} = 0$$

$$\times 20k: 2(2) + 2 - V_1 = 0$$

$$\therefore V_1 = 6V$$

Question 12 [2]

Vraag 12 [2]

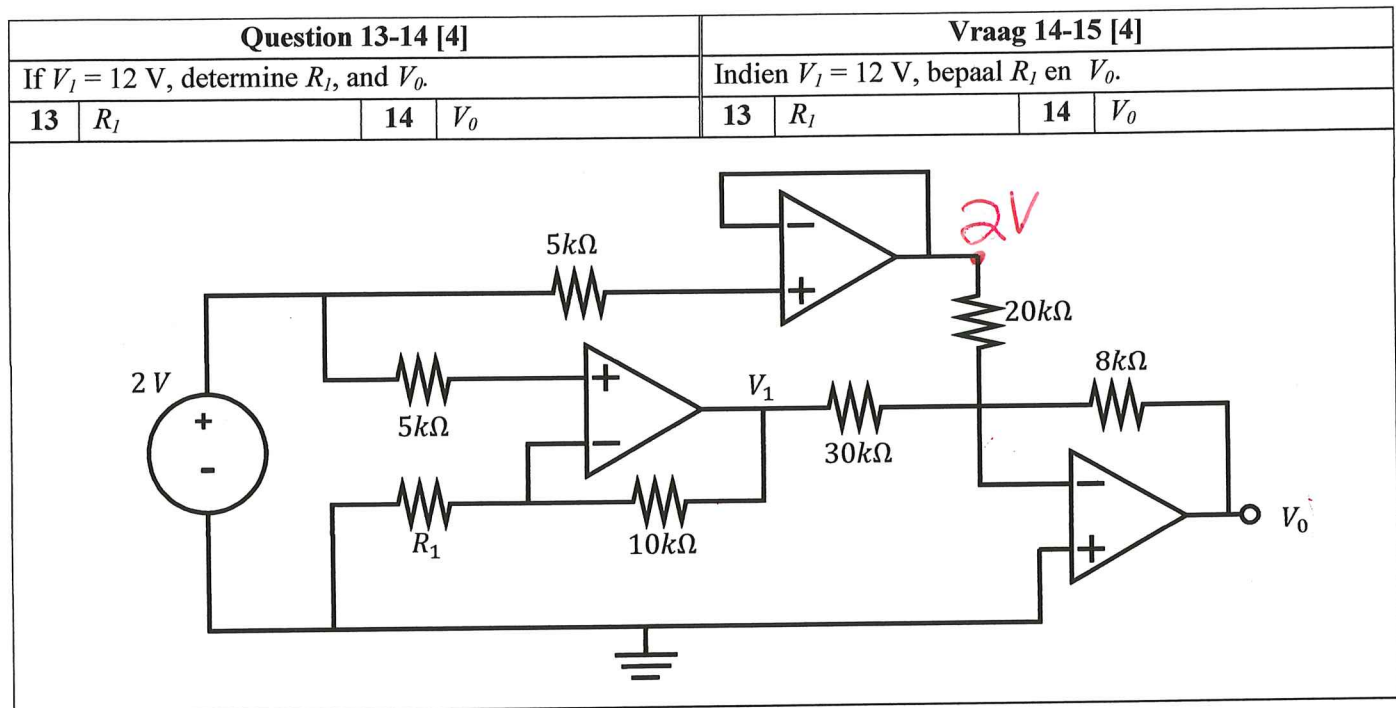
12 Determine v_2 .12 Bepaal v_2 .

$$\frac{3 - V_2}{4k} + \frac{3}{8k} = 0,5m$$

$$\times 8k: 6 - 2V_2 + 3 = 4$$

$$\therefore 2V_2 = 5$$

$$V_2 = 2,5V$$



$$V_1 = 12 = \left(\frac{10k}{R_I} + 1 \right) (2)$$

$$6 = \frac{10k}{R_I} + 1$$

$$\therefore R_I = 2k\Omega$$

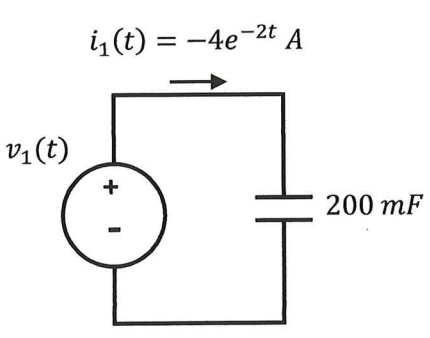
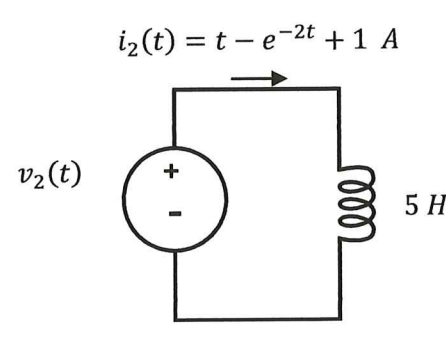
$$\frac{0 - V_1}{30k} + \frac{0 - 2}{20k} + \frac{0 - V_O}{8k} = 0$$

$$\times 60k: -2(12) - 3(2) - 7.5V_O = 0$$

$$\therefore 7.5V_O = -30$$

$$V_O = -4\text{ V}$$

SECTION / AFDELING 3 [8]

Question 15-18 [4]				Vraag 15-18 [4]			
Determine $v_1(t)$ and $v_2(t)$ and put them in the following form: $v_1(t) = ae^{-2t} + b$ $v_2(t) = ce^{-2t} + d$ Assume $v_1(0) = 10\text{ V}$ and $v_2(0) = 15\text{ V}$.				Bepaal $v_1(t)$ en $v_2(t)$ en skryf die antwoorde in die volgende vorm: $v_1(t) = ae^{-2t} + b$ $v_2(t) = ce^{-2t} + d$ Aanvaar $v_1(0) = 10\text{ V}$ en $v_2(0) = 15\text{ V}$.			
15	a	16	b	15	a	16	b
17	c	18	d	17	c	18	d
$i_1(t) = -4e^{-2t}\text{ A}$ 				$i_2(t) = t - e^{-2t} + 1\text{ A}$ 			

$$V_1 = \frac{1}{C} \int_0^t -4e^{-2t} dt + 10$$

$$= -\frac{4}{200\text{m}} \left[-\frac{1}{2} e^{-2t} \right]_0^t + 10$$

$$= 10 [e^{-2t} - 1] + 10$$

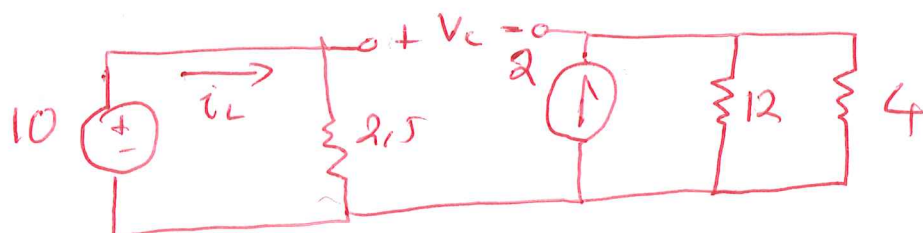
$$V_1 = 10e^{-2t} \text{ V} : \underline{a=10; b=0}$$

$$V_2 = 5 \frac{d}{dt} (t - e^{-2t} + 1)$$

$$= 5(1 + 2e^{-2t})$$

$$V_2 = 10e^{-2t} + 5 \text{ V} : \underline{c=10; d=5}$$

Question 19 & 20 [4]				Vraag 19 & 20 [4]			
Assume DC conditions and determine the energy stored in the capacitor (w_C) and the inductor (w_L).				Aanvaar gelykstroom toestande en bepaal die energie gestoor in die kapasitor (w_C) en die inductor (w_L).			
19	w_C	20	w_L	19	w_C	20	w_L



$$i_L = \frac{10}{2,5} = 4 \text{ A}$$

$$w_L = \frac{1}{2} (150 \text{ m})(4^2) \quad \text{no}$$

$$\underline{w_L = 1,2 \text{ J}}$$

$$V_C = 10 - 2(12//4)$$

$$= 10 - 6 = 4 \text{ V}$$

$$\therefore w_C = \frac{1}{2} (25 \text{ m})(4^2)$$

$$\underline{w_C = 0,2 \text{ J}}$$

SECTION / AFDELING 4 [12]

Question 21-24 [4]				Vraag 21-24 [4]			
a) Find r and θ in the equation below $r\angle\theta = \frac{(10 + j20)^2}{\sqrt{(10 + j5)(16 - j20)}}$				a) Bereken r en θ vir onderstaande vergelyking $r\angle\theta = \frac{(10 + j20)^2}{\sqrt{(10 + j5)(16 - j20)}}$			
b) If the determinant of the matrix is $\Delta = 46 + j24$, find a and b . $\begin{bmatrix} 2 + j3 & -j2 \\ -j2 & a + jb \end{bmatrix}$				b) Indien die determinant van onderstaande matriks $\Delta = 46 + j24$ is, bereken a en b . $\begin{bmatrix} 2 + j3 & -j2 \\ -j2 & a + jb \end{bmatrix}$			
21	r	22	θ	21	r	22	θ
23	a	24	b	23	a	24	b

$$\begin{aligned}
 a) \quad r\angle\theta &= \frac{-300 + j400}{\sqrt{286,35642 - 24,77514}} \\
 &= \frac{-300 + j400}{16,92207 - 12,38757} \\
 r\angle\theta &= 29,54721 / 139,25747 \rightarrow
 \end{aligned}$$

$$b) \quad \Delta = 46 + j24 = \begin{vmatrix} 2 + j3 & -j2 \\ -j2 & a + jb \end{vmatrix}$$

$$46 + j24 = (2 + j3)(a + jb) - (-j2)(-j2)$$

$$42 + j24 = (2 + j3)(a + jb)$$

$$\therefore a + jb = 12 - 6j \rightarrow$$

Question 25 - 28 [4]				Vraag 25 - 28 [4]			
For the signals below: $i_1(t) = -2 \cos(\omega t + 130^\circ) \text{ A}$ $i_2(t) = 2 \sin(\omega t + 160^\circ) \text{ A}$ a) Which signal is leading (L), and by how many degrees (Φ)? (L = 1 if $i_1(t)$ is leading and L = 2 if $i_2(t)$ is leading) b) Find $i_3(t) = i_1(t) + i_2(t)$ and write it in the following form: $i_3(t) = I_m \cos(\omega t + \theta)$ with $I_m > 0$ and $ \theta \leq 180^\circ$.				Vir onderstaande seine: $i_1(t) = -2 \cos(\omega t + 130^\circ) \text{ A}$ $i_2(t) = 2 \sin(\omega t + 160^\circ) \text{ A}$ a) Watter sein is voorlopend (L), en moet hoeveel grade (Φ)? (L = 1 if $i_1(t)$ is leading and L = 2 if $i_2(t)$ is leading) b) Bereken $i_3(t) = i_1(t) + i_2(t)$ en skryf die antwoord in die volgende vorm: $i_3(t) = I_m \cos(\omega t + \theta)$ met $I_m > 0$ en $ \theta \leq 180^\circ$.			
25	L	26	Φ	25	L	26	Φ
27	I_m	28	θ	27	I_m	28	θ

$$i_1(t) = 2 \cos(\omega t + 130 - 180) \\ = 2 \cos(\omega t - 50^\circ) \text{ A}$$

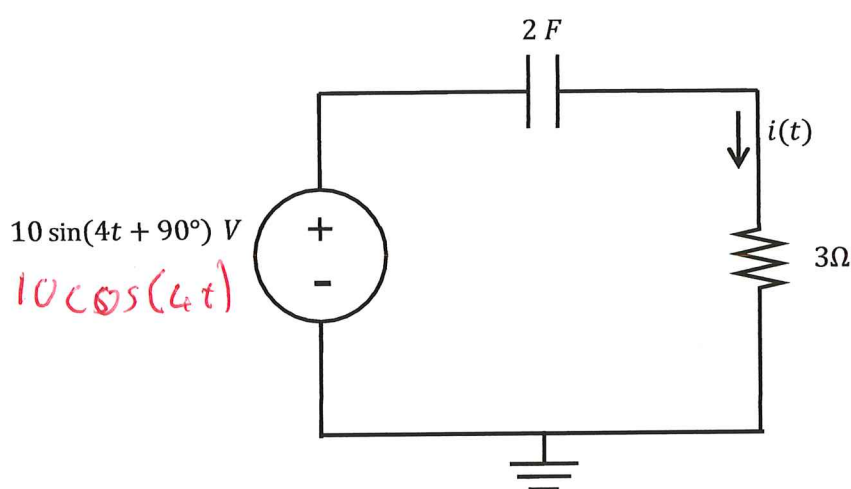
$$i_2(t) = 2 \cos(\omega t + 160^\circ - 90^\circ) \\ = 2 \cos(\omega t + 70^\circ)$$

a) i_2 leads by $\Phi = 70 + 50 = 120^\circ$

$$b) i_3 = 2 \angle -50^\circ + 2 \angle 70^\circ \\ = 2 \angle 10^\circ$$

$$\therefore i_3(t) = 2 \cos(\omega t + 10^\circ)$$

Question 29 - 32 [4]				Vraag 29 - 32 [4]			
a) A linear circuit has a current input of $i(t) = 7.5 \cos(10t + 30)$ A and a voltage output of $v(t) = 120 \cos(10t + 75)$ V. Determine the impedance of the circuit as a phasor in rectangular form ($a + jb$) . (Impedance: voltage phasor divided by current phasor.) b) Find $i(t)$ in the circuit below and write it in the following form: $i(t) = I_m \cos(4t + \phi)$ with $I_m > 0$ and $ \phi \leq 180^\circ$.				a) 'n Linêre kringbaan het 'n stroom inset van $i(t) = 7.5 \cos(10t + 30)$ A en 'n spannings uitset $v(t) = 120 \cos(10t + 75)$ V. Bepaal die impedansie van die kringbaan in reghoekige format ($a + jb$) . (Impedansie: spanningsfasor gedeel deur stroomfasor.) b) Bepaal $i(t)$ vir onderstaande kringbaan en skryf in die volgende vorm: $i(t) = I_m \cos(4t + \phi)$ met $I_m > 0$ en $ \phi \leq 180^\circ$.			
29	a	30	b	29	a	30	b
31	I_m	32	ϕ	31	I_m	32	ϕ



$$a) Z = \frac{120 \angle 75}{7.5 \angle 30} = 11.31371 + j11.31371$$

$$b) Z = 3 + \frac{1}{j(4)(2)} = 3 - j0.125 = 3.0026 \angle -2.38594$$

$$I = \frac{10 \angle 0}{3.0026 \angle -2.38594}$$

$$= 3.33067 \angle 2.38594$$

$$i(t) = 3.33067 \cos(4t + 2.38594)$$

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PRACTICE PAGE / OEFENBLAD 1

Student number

M E M O

Assessment ID

2 0 1 7 E B N 1 1 1 S 2

Make use of 5 significant digits for all answers involving irrational numbers.

Maak gebruik van 5 beduidende syfers vir alle antwoorde bestaande uit irrasionele getalle.

Answer

Unit

01	$V_{TH} =$	40	V
02	$R_{TH} =$	500	Ω
03	$V_{TH} =$	33	V
04	$R_{TH} =$	45	$k\Omega$
05	$I_N =$	-200	mA
06	$R_N =$	200	Ω
07	$V_0 =$	-36	V
08	$V_0 =$	-9	V
09	$a =$	0	$k\Omega$
10	$b =$	1	$k\Omega$
11	$v_1 =$	6	V
12	$v_2 =$	2,5	V
13	$R_I =$	2	$k\Omega$
14	$V_0 =$	-4	V
15	$a =$	10	
16	$b =$	0	
17	$c =$	10	
18	$d =$	5	
19	$w_C =$	0,2	J
20	$w_L =$	1,2	J
21	$r =$	29,54721	
22	$\theta =$	139,25747	°
23	$a =$	12	
24	$b =$	-6	

PRACTICE PAGE / OEFENBLAD 2

Student number	M	E	M	O										
Assessment ID	2	0	1	7	E	B	N	1	1	1	S	2		

Make use of 5 significant digits for all answers involving irrational numbers.
 Maak gebruik van 5 beduidende syfers vir alle antwoorde bestaande uit irrasionele getalle.

	Answer	Unit
25	$L = 2$	
26	$\Phi = 120^\circ$	°
27	$I_m = 2$	A
28	$\theta = 10$	°
29	$a = 11,31371$	
30	$b = 11,31371$	
31	$A = 3,33067$	A
32	$\phi = 2,38594$	°